

2.20 解答:

(a)四根无限长线电荷位于同一圆周上, 半径为 a , 角位置分别为

$$\phi_n = \frac{n\pi}{2}, \quad n = 0, 1, 2, 3.$$

电荷线密度交替为 $+\lambda, -\lambda, +\lambda, -\lambda$, 因此二维电荷密度可以写成

$$\sigma(\rho, \phi) = \frac{\lambda}{a} \sum_{n=0}^3 (-1)^n \delta(\rho - a) \delta\left(\phi - \frac{n\pi}{2}\right).$$

二维自由空间 Green 函数在极坐标中的展开式为

$$G(\rho, \phi; \rho', \phi') = -\frac{1}{2\pi} \ln \rho_{>} + \frac{1}{2\pi} \sum_{m=1}^{\infty} \frac{1}{m} \left(\frac{\rho_{<}}{\rho_{>}}\right)^m \cos m(\phi - \phi'),$$

其中

$$\rho_{<} = \min(\rho, \rho'), \quad \rho_{>} = \max(\rho, \rho').$$

静电势由

$$\Phi(\rho, \phi) = \frac{1}{\epsilon_0} \int G(\rho, \phi; \rho', \phi') \sigma(\rho', \phi') \rho' d\rho' d\phi'$$

给出。代入上面的线电荷密度, 得

$$\Phi(\rho, \phi) = \frac{\lambda}{\epsilon_0} \sum_{n=0}^3 (-1)^n G\left(\rho, \phi; a, \frac{n\pi}{2}\right).$$

$$\Phi(\rho, \phi) = \frac{\lambda}{2\pi\epsilon_0} \sum_{m=1}^{\infty} \frac{1}{m} \left(\frac{\rho_{<}}{\rho_{>}}\right)^m \sum_{n=0}^3 (-1)^n \cos m\left(\phi - \frac{n\pi}{2}\right).$$

现在计算角向求和。令

$$S_m = \sum_{n=0}^3 (-1)^n \cos m\left(\phi - \frac{n\pi}{2}\right).$$

用复指数表示,

$$S_m = \operatorname{Re} \left[e^{im\phi} \sum_{n=0}^3 (-1)^n e^{-imn\pi/2} \right].$$

括号中的求和是几何级数:

$$\sum_{n=0}^3 \left(-e^{-im\pi/2}\right)^n.$$

只有当

$$-e^{-im\pi/2} = 1$$

时它不为零。也就是说

$$e^{-im\pi/2} = -1,$$

因此

$$m = 4k + 2, \quad k = 0, 1, 2, \dots$$

在这种情况下,

$$S_m = 4 \cos m\phi.$$

因此只有 $m = 4k + 2$ 的项留下, 电势为

$$\Phi(\rho, \phi) = \frac{\lambda}{2\pi\epsilon_0} \sum_{k=0}^{\infty} \frac{1}{4k+2} \left(\frac{\rho_{<}}{\rho_{>}} \right)^{4k+2} 4 \cos[(4k+2)\phi].$$

整理得

$$\Phi(\rho, \phi) = \frac{\lambda}{\pi\epsilon_0} \sum_{k=0}^{\infty} \frac{1}{2k+1} \left(\frac{\rho_{<}}{\rho_{>}} \right)^{4k+2} \cos[(4k+2)\phi].$$

其中这里

$$\rho_{<} = \min(\rho, a), \quad \rho_{>} = \max(\rho, a).$$

特别地, 在四根线电荷围成的内部区域 $\rho < a$ 中, 有

$$\rho_{<} = \rho, \quad \rho_{>} = a,$$

所以

$$\Phi(\rho, \phi) = \frac{\lambda}{\pi\epsilon_0} \sum_{k=0}^{\infty} \frac{1}{2k+1} \left(\frac{\rho}{a} \right)^{4k+2} \cos[(4k+2)\phi].$$

(b)题中给出的复函数为

$$w(z) = \frac{2\lambda}{4\pi\epsilon_0} \ln \left[\frac{(z - ia)(z + ia)}{(z - a)(z + a)} \right], \quad z = x + iy = \rho e^{i\phi}.$$

也就是

$$w(z) = \frac{\lambda}{2\pi\epsilon_0} \ln \left[\frac{z^2 + a^2}{z^2 - a^2} \right].$$

静电势应当取这个复函数的实部:

$$\Phi(\rho, \phi) = \operatorname{Re} w(z).$$

先考虑四根线电荷围成的内部区域 $\rho < a$ 。令

$$u = \frac{z^2}{a^2}.$$

则

$$\frac{z^2 + a^2}{z^2 - a^2} = -\frac{1 + u}{1 - u}.$$

因此

$$w(z) = \frac{\lambda}{2\pi\epsilon_0} [\ln(-1) + \ln(1+u) - \ln(1-u)].$$

其中 $\ln(-1)$ 只贡献一个虚常数，对实部没有影响。所以

$$\operatorname{Re} w(z) = \frac{\lambda}{2\pi\epsilon_0} \operatorname{Re} [\ln(1+u) - \ln(1-u)].$$

利用展开式

$$\ln(1+u) - \ln(1-u) = 2 \left(u + \frac{u^3}{3} + \frac{u^5}{5} + \cdots \right), \quad |u| < 1,$$

得到

$$\operatorname{Re} w(z) = \frac{\lambda}{\pi\epsilon_0} \sum_{k=0}^{\infty} \frac{1}{2k+1} \operatorname{Re} [u^{2k+1}].$$

由于

$$u^{2k+1} = \left(\frac{z^2}{a^2} \right)^{2k+1} = \left(\frac{\rho}{a} \right)^{4k+2} e^{i(4k+2)\phi},$$

所以

$$\operatorname{Re} [u^{2k+1}] = \left(\frac{\rho}{a} \right)^{4k+2} \cos[(4k+2)\phi].$$

因此

$$\boxed{\Phi(\rho, \phi) = \operatorname{Re} w(z) = \frac{\lambda}{\pi\epsilon_0} \sum_{k=0}^{\infty} \frac{1}{2k+1} \left(\frac{\rho}{a} \right)^{4k+2} \cos[(4k+2)\phi]}, \quad \rho < a.$$

这正是(a)中在内部区域得到的结果。

对于外部区域 $\rho > a$ ，可以令

$$v = \frac{a^2}{z^2}.$$

则

$$\frac{z^2 + a^2}{z^2 - a^2} = \frac{1+v}{1-v}.$$

所以

$$w(z) = \frac{\lambda}{2\pi\epsilon_0} [\ln(1+v) - \ln(1-v)].$$

同样展开得

$$\operatorname{Re} w(z) = \frac{\lambda}{\pi\epsilon_0} \sum_{k=0}^{\infty} \frac{1}{2k+1} \left(\frac{a}{\rho} \right)^{4k+2} \cos[(4k+2)\phi], \quad \rho > a.$$

因此内外两侧可以统一写成

$$\Phi(\rho, \phi) = \frac{\lambda}{\pi\epsilon_0} \sum_{k=0}^{\infty} \frac{1}{2k+1} \left(\frac{\rho_{<}}{\rho_{>}} \right)^{4k+2} \cos[(4k+2)\phi],$$

其中

$$\rho_{<} = \min(\rho, a), \quad \rho_{>} = \max(\rho, a).$$

这与(a)中用 Green 函数展开得到的结果一致。

(c) 保留 $k=0$ 和 $k=1$ 两项，有

$$\Phi(\rho, \phi) \simeq \frac{\lambda}{\pi\epsilon_0} \left[\left(\frac{\rho}{a} \right)^2 \cos 2\phi + \frac{1}{3} \left(\frac{\rho}{a} \right)^6 \cos 6\phi \right].$$

利用

$$\rho^2 \cos 2\phi = x^2 - y^2,$$

以及

$$\rho^6 \cos 6\phi = \operatorname{Re}(x + iy)^6 = x^6 - 15x^4y^2 + 15x^2y^4 - y^6,$$

得到

$$\Phi(x, y) \simeq \frac{\lambda}{\pi\epsilon_0} \left[\frac{x^2 - y^2}{a^2} + \frac{x^6 - 15x^4y^2 + 15x^2y^4 - y^6}{3a^6} \right].$$

因此

$$\begin{aligned} E_x &= -\frac{\partial\Phi}{\partial x} = -\frac{2\lambda}{\pi\epsilon_0} \left[\frac{x}{a^2} + \frac{x^5 - 10x^3y^2 + 5xy^4}{a^6} \right], \\ E_y &= -\frac{\partial\Phi}{\partial y} = \frac{2\lambda}{\pi\epsilon_0} \left[\frac{y}{a^2} + \frac{5x^4y - 10x^2y^3 + y^5}{a^6} \right]. \end{aligned}$$

当 $y=0$ 时，

$$E_x = -\frac{2\lambda}{\pi\epsilon_0} \left[\frac{x}{a^2} + \frac{x^5}{a^6} \right].$$

所以

$$E_x^{(0)} = -\frac{2\lambda}{\pi\epsilon_0} \frac{x}{a^2}, \quad E_x^{(1)} = -\frac{2\lambda}{\pi\epsilon_0} \frac{x^5}{a^6}.$$

因此相对大小为

$$\left| \frac{E_x^{(1)}}{E_x^{(0)}} \right| = \left(\frac{x}{a} \right)^4.$$